DATA ANALYST TRAINING ROADMAP - 2025

Start your career to Data Analyst, this well-designed course will help you to learn Data Analyst skills from scratch and covering both foundational and advanced topics:

Role of a Data Analyst

A Data Analyst collects, cleans, and interprets data sets in order to answer a question or solve a problem

Six steps of Data Analysis:

1. **Business Question:** Define what problem you want to solve
2. **Get Data:** Collect the data required for analysis
3. **Explore Data:** Explore data with visual exploration to understand what is in a dataset
4. **Prepare Data:** Data cleaning, calculated fields and data validation
5. **Analyze Data:** Use data analysis techniques to understand, interpret, and derive conclusions based on the requirements
6. **Present Findings:** Share insights with stakeholders

Module 1: Programming and Data Analytics with Python

Python is a versatile language widely used for data analysis. This course will equip you with the skills to use Python for data manipulation, visualization, and statistical analysis.

Key Learning Outcomes:

- Learn the Python programming language for data analysis.
- Utilize Python libraries like NumPy, Pandas, and Matplotlib for data manipulation and visualization.
- Apply statistical and machine learning techniques using Python.
- Develop data-driven applications and solutions.

Course Curriculum:

Introduction and Basics

- Installation
- Python Org, Python 3
- Variables
- Print function
- Input from user
- Data Types
- Type Conversion
- First Program

Operators

- Arithmetic Operators
- Relational Operators
- Bitwise Operators
- Logical Operators
- Assignment Operators
- Compound Operators
- Membership Operators
- Identity Operators

Conditional Statements

- If Else
- If
- Else
- El If (else if)
- If Else Ternary Expression

While Loop

- While loop logic building
- Series based Questions
- Break
- Continue
- Nested While Loops
- Pattern-Based Questions
- pass
- Loop else

Lists

- List Basics
- List Operations
- List Comprehensions / Slicing
- List Methods

Strings

- String Basics
- String Literals
- String Operations
- String Comprehensions / Slicing
- String Methods

For Loops

- Range function
- For loop
- Nested For Loops
- Pattern-Based Questions

- Break
- Continue
- Pass
- Loop else

Functions

- Definition
- Call
- Function Arguments
- Default Arguments
- Docstrings
- Scope
- Special functions Lambda, Map, and Filter
- Recursion
- Functional Programming and Reference Functions

Dictionary

- Dictionaries Basics
- Operations
- Comprehensions
- Dictionaries Methods

Tuple

- Tuples Basics
- Tuples Comprehensions / Slicing
- Tuple Functions
- Tuple Methods

Set

- Sets Basics
- Sets Operations
- Union
- Intersection
- Difference and Symmetric Difference

Object-Oriented Programming

- Classes
- Objects
- Method Calls
- Inheritance and Its Types
- Overloading
- Overriding
- Data Hiding
- Operator Overloading

File Handling

- File Basics
- Opening Files
- Reading Files
- Writing Files
- Editing Files
- Working with different extensions of file
- With Statements

Exception Handling

- Common Exceptions
- Exception Handling
- Try
- Except
- Try except else
- Finally
- Raising exceptions
- Assertion

Regular Expression

- Basic RE functions
- Patterns
- Meta Characters
- Character Classes

Modules & Packages

- Different types of modules
- Inbuilt modules
- OS
- Sys
- Statistics
- Math
- String
- Random
- Create your own module
- Building Packages
- Build your own python module and deploy it on pip

Data Structures

- Stack
- Queue
- Linked Lists
- Sorting
- Searching
- Linear Search
- Binary Search

Higher-Order Functions

- Function as a parameter
- Function as a return value
- Closures
- Decorators
- Map, Filter, Reduce Functions

Python Web Scrapping

- Understanding BeautifulSoup
- Extracting Data from websites
- Extracting Tables
- Data in JSON format

Python Package Manager

- What is PIP?
- Installation
- PIP Freeze
- Creating Your Own Package
- Upload it on PIP

Python with MongoDB Database

- SQL and NoSQL
- Connecting to MongoDB URI
- Flask application and MongoDB integration
- CRUD Operations
- Find
- Delete
- Drop

Building API

- API (Application Programming Interface)
- Building API
- Structure of an API
- PUT
- POST
- DELETE
- Using Postman

Data Analysis with Python

Understanding NumPy

NumPy basics

- Working with Matrix
- Linear Algebra operations

Statistics

- Descriptive Statistics
- Normal Distribution Operations
- Mean, Variance and Standard Deviation

- Reshaping arrays

Exploratory Data Analysis (EDA) with Pandas

Pandas

- Data Analysis basics
- Dataframe operations
- Working with 2-dimensional data
- Data Cleaning
- Data Grouping

Working with Datasets

Data Visualization with Matplotlib and Seaborn

Matplotlib

- Plot Basics
- Format Strings
- Label and Legends
- Bar Chart
- Pie Chart

Module 2: Statistical Analysis and Data Analytics Projects

1. Statistics and Statistical Models

Statistics

Descriptive Statistics

- Measure of Frequency and Central Tendency
- Measure of Dispersion
- Probability Distribution
- Gaussian Normal Distribution
- Skewness and Kurtosis
- Regression Analysis
- Continuous and Discrete Functions
- Goodness of Fit
- Normality Test
- ANOVA
- Homoscedasticity
- Linear and Non-Linear Relationship with Regression

Statistical Models

Inferential Statistics

- t-Test
- z-Test
- Hypothesis Testing
- Type I and Type II errors
- t-Test and its types
- One way ANOVA
- Two way ANOVA
- Chi-Square Test
- Implementation of continuous and categorical data

2. Working with Different Types of Datasets – Projects

Exploratory Data Analysis Project

Module 3: SQL Fundamentals

SQL is a powerful language for interacting with databases and extracting valuable information. This course will teach you the fundamentals of SQL syntax, database design, and query writing.

Key Learning Outcomes:

- Master the fundamentals of SQL syntax and database management.
- Query databases efficiently to extract relevant data.
- Understand database design concepts and normalization.
- Work with different database management systems (DBMS).

Course Outline:

1. Introduction to SQL
2. Data types and operators
3. Database design
4. SQL queries (SELECT, INSERT, UPDATE, DELETE)
5. Advanced SQL concepts (joins, subqueries, window functions)

Module 4: Business Analytics Using Excel

This course provides a solid foundation in business analytics using Excel, a widely used tool for data analysis. You'll learn how to clean, prepare, and analyze data using Excel's functions and tools, as well as create effective data visualizations to communicate insights.

Key Learning Outcomes:

- Develop proficiency in using Excel for data analysis and visualization.
- Learn to apply statistical and financial functions to analyze data.
- Create effective data visualizations using Excel's charting tools.
- Understand basic business analytics concepts and metrics.

Course Outline:

1. Data cleaning and preparation
2. Descriptive statistics
3. Data visualization techniques
4. Financial analysis
5. Case studies in business analytics

Module 5: Power BI

Power BI is another popular tool for data visualization and reporting. This course will teach you how to use Power BI to create interactive dashboards and analyze data.

Key Learning Outcomes:

- Learn to use Power BI for data visualization and reporting.
- Connect to various data sources and create interactive dashboards.
- Use Power BI's features for data modeling and analysis.
- Share and collaborate on Power BI reports.

Course Curriculum:

Introduction To Power BI

- Introduction to Power BI – Need, Importance
- Power BI – Advantages and Scalable Options

- History – Power View, Power Query, Power Pivot
- Power BI Data Source Library and DW Files
- Cloud Collaboration and Usage Scope
- Business Analyst Tools, MS Cloud Tools
- Power BI Installation and Cloud Account
- Power BI Cloud and Power BI Service
- Power BI Architecture and Data Access
- On-Premise Data Access and Microsoft on Drive
- Power BI Desktop – Installation, Usage
- Sample Reports and Visualization Controls
- Power BI Cloud Account Configuration
- Understanding Desktop & Mobile Editions
- Report Rendering Options and End User Access
- Power View and Power Map. Power BI Licenses

Creating POWER BI Reports, Auto Filters

- Report Design with Legacy & .DAT Files
- Report Design with Data Base Tables
- Understanding Power BI Report Designer
- Report Canvas, Report Pages: Creation, Renames
- Report Visuals, Fields and UI Options
- Experimenting Visual Interactions, Advantages
- Reports with Multiple Pages and Advantages
- Pages with Multiple Visualizations. Data Access
- PUBLISH Options and Report Verification in Cloud
- "GET DATA" Options and Report Fields, Filters
- Report View Options: Full, Fit Page, Width Scale
- Report Design using Databases & Queries
- Query Settings and Data Preloads
- Navigation Options and Report Refresh
- Stacked bar chart, Stacked column chart

- Clustered bar chart, Clustered column chart
- Adding Report Titles. Report Format Options
- Focus Mode, Explore and Export Settings

Report Visualization and Properties

- Power BI Design: Canvas, Visualizations and Fields
- Import Data Options with Power BI Model, Advantages
- Direct Query Options and Real-time (LIVE) Data Access
- Data Fields and Filters with Visualizations
- Visualization Filters, Page Filters, Report Filters
- Conditional Filters and Clearing. Testing Sets
- Creating Customised Tables with Power BI Editor
- General Properties, Sizing, Dimensions, and Positions
- Alternate Text and Tiles. Header (Column, Row) Properties
- Grid Properties (Vertical, Horizontal) and Styles
- Table Styles & Alternate Row Colors – Static, Dynamic
- Sparse, Flashy Rows, Condensed Table Reports. Focus Mode
- Totals Computations, Background. Boders Properties
- Column Headers, Column Formatting, Value Properties
- Conditional Formatting Options – Color Scale
- Page Level Filters and Report Level Filters
- Visual-Level Filters and Format Options
- Report Fields, Formats and Analytics
- Page-Level Filters and Column Formatting, Filters
- Background Properties, Borders and Lock Aspect

Chart And Map Report Properties

- CHART Report Types and Properties
- STACKED BAR CHART, STACKED COLUMN CHART
- CLUSTERED BAR CHART, CLUSTERED COLUMN CHART
- 100% STACKED BAR CHART, 100% STACKED COLUMN CHART

- LINE CHARTS, AREA CHARTS, STACKED AREA CHARTS
- LINE AND STACKED ROW CHARTS
- LINE AND STACKED COLUMN CHARTS
- WATERFALL CHART, SCATTER CHART, PIE CHART
- Field Properties: Axis, Legend, Value, Tooltip
- Field Properties: Color Saturation, Filters Types
- Formats: Legend, Axis, Data Labels, Plot Area
- Data Labels: Visibility, Color and Display Units
- Data Labels: Precision, Position, Text Options
- Analytics: Constant Line, Position, Labels
- Working with Waterfall Charts and Default Values
- Modifying Legends and Visual Filters – Options
- Map Reports: Working with Map Reports
- Hierarchies: Grouping Multiple Report Fields
- Hierarchy Levels and Usages in Visualizations
- Preordered Attribute Collection – Advantages
- Using Field Hierarchies with Chart Reports
- Advanced Query Mode @ Connection Settings – Options
- Direct Import and In-memory Loads, Advantages

Hierarchies And Drilldown Reports

- Hierarchies and Drilldown Options
- Hierarchy Levels and Drill Modes – Usage
- Drill-thru Options with Tree Map and Pie Chart
- Higher Levels and Next Level Navigation Options
- Aggregates with Bottom/Up Navigations. Rules
- Multi Field Aggregations and Hierarchies in Power BI
- DRILLDOWN, SHOWNEXTLEVEL, EXPANDTONEXTLEVEL
- SEE DATA and SEE RECORDS Options. Differences
- Toggle Options with Tabular Data. Filters
- Drilldown Buttons and Mouse Hover Options @ Visuals

- Dependant Aggregations, Independent Aggregations
- Automated Records Selection with Tabular Data
- Report Parameters: Creation and Data Type
- Available Values and Default values. Member Values
- Parameters for Column Data and Table / Query Filters
- Parameters Creation – Query Mode, UI Option
- Linking Parameters to Query Columns – Options
- Edit Query Options and Parameter Manage Entries
- Connection Parameters and Dynamic Data Sources
- Synonyms – Creation and Usage Options

Power Query & M Language

- Understanding Power Query Editor – Options
- Power BI Interface and Query / Dataset Edits
- Working with Empty Tables and Load / Edits
- Empty Table Names and Header Row Promotions
- Undo Headers Options. Blank Columns Detection
- Data Imports and Query Marking in Query Editor
- JSON Files & Binary Formats with Power Query
- JavaScript Object Notation – Usage with M Lang.
- Applied Steps and Usage Options. Revert Options
- creating Query Groups and Query References. Usage
- Query Rename, Load Enable and Data Refresh Options
- Combine Queries – Merge Join and Anti-Join Options
- Combine Queries – Union and Union All as New Dataset
- M Language: NestedJoin and JoinKind Functions
- REPLACE, REMOVE ROWS, REMOVE COL, BLANK – M Lang
- Column Splits and FilledUp / FilledDown Options
- Query Hide and Change Type Options. Code Generation

DAX EXPRESSIONS – Level 1

- Purpose of Data Analysis Expressions (DAX)
- Scope of Usage with DAX. Usability Options
- DAX Context: Row Context and Filter Context
- DAX Entities: Calculated Columns and Measures
- DAX Data Types: Numeric, Boolean, Variant, Currency
- Datetime Data Tye with DAX. Comparison with Excel
- DAX Operators & Symbols. Usage. Operator Priority
- Parenthesis, Comparison, Arithmetic, Text, Logic
- DAX Functions and Types: Table Valued Functions
- Filter, Aggregation and Time Intelligence Functions
- Information Functions, Logical, Parent-Child Functions
- Statistical and Text Functions. Formulas and Queries
- Syntax Requirements with DAX. Differences with Excel
- Naming Conventions and DAX Format Representation
- Working with Special Characters in Table Names
- Attribute / Column Scope with DAX – Examples
- Measure / Column Scope with DAX – Examples

DAX EXPRESSIONS – Level 2

- YTD, QTD, MTD Calculations with DAX
- DAX Calculations and Measures
- Using TOPN, RANKX, RANK.EQ
- Computations using STDEV & VAR
- SAMPLE Function, COUNTALL, ISERROR
- ISTEXT, DATEFORMAT, TIMEFORMAT
- Time Intelligence Functions with DAX
- Data Analysis Expressions and Functions
- DATESYTD, DATESQTD, DATESMTD
- ENDOFYEAR, ENDOFQUARTER,ENDOFMONTH
- FIRSTDATE, LASTDATE, DATESBETWEEN
- CLOSINGBALANCEYEAR,CLOSINGBALANCEQTR

- SAMEPERIOD and PREVIOUSMONTH,QUARTER
- KPIs with DAX. Vertipaq Queries in DAX
- IF..ELSEIF.. Conditions with DAX
- Slicing and Dicing Options with Columns, Measures
- DAX for Query Extraction, Data Mashup Operations
- Calculated Columns and Calculated Measures with DAX

Power BI Deployment & Cloud

- Power BI Report Validation and Publish
- Understanding Power BI Cloud Architecture
- Power BI Cloud Account and Workspace
- Reports and Datasets Items Validation
- Dashboards and Pins – Real-time Usage
- Dynamic Data Sources and Encryptions
- Personal and Organizational Content Packs
- Gateways, Subscriptions, Mobile Reports
- Data Refresh with Power BI Architecture
- PBIX and PBIT Files with Power BI – Usage
- Visual Data Imparts and Visual Schemas
- Cloud and On-Premise Data Sources
- How Power BI Supports Data Model?
- Relation between Dashboards to Reports
- Relation between Datasets to Reports
- Relation between Datasets to Dashboards
- Page to Report – Mapping Options
- Publish Options and Data Import Options
- Need for PINS @ Visuals and PINS @ Reports
- Need for Data Streams and Cloud Integration

Power BI Cloud Operations

- Report Publish Options and Verifications
- Working with Power BI Cloud Interface & Options

- Navigation Paths with “My Workspace” Screens
- FILE, VIEW, EDIT REPORTS, ACCESS, DRILLDOWN
- Saving Reports into pdf, pptx, etc. Report Embed
- Report Rendering and EDIT, SAVE, Print Options
- Report PIN and individual Visual PIN Options
- Create and Use Dashboards. Menu Options
- Goto Dashboard and Goto LIVE Page Options
- Operations on Pinned Reports and Visuals
- TITLE, MEDIA, USAGE METRICS & FAVOURITES
- SUBSCRIPTION Options and Reports with Mobile View
- Options with Report Page : Print and Subscribe
- Report Actions: USAGE METRICS, ANALYSE IN EXCEL
- Report Actions: RELATED ITEMS, RENAME, DELETE
- Dashboard Actions: METRICS, RELATED ITEMS
- Dashboard Actions: SETTINGS FOR Q & A, DELETE
- PIN Actions: METRICS, SHARE, RELATED ITEMS
- PIN Actions: SETTINGS FOR Q & A, DELETE
- EDIT DASHBOARD (CLOUD), On-The-Fly Reports
- Dataset Actions: CREATE REPORT, REFRESH
- SCHEDULED REFRESH & RELATED ITEMS
- Dashboard Integration with Apps in Power BI

Improving Power BI Reports

- Publish Power BI Report Templates
- Import and Export Options with Power BI
- Dataset Navigations and Report Navigations
- Quick Navigation Options with “My Workspace”
- Dashboards, Workbooks, Reports, Datasets
- Working with MY WORK SPACE group
- Installing the Power BI Personal Gateway
- Automatic Refresh – Possible Issues

- Adding images to the dashboards
- Reading & Editing Power BI Views
- Power BI Templates (pbit)- Creation, Usage
- Managing report in Power BI Services
- Power BI Gateway – Download and Installation
- Personal and Enterprise Gateway Features
- Power BI Settings: Dataset – Gateway Integration
- Configuring Dataset for Manual Refresh of Data
- Configuring Automatic Refresh and Schedules
- Workbooks and Alerts with Power BI
- Dataset Actions and Refresh Settings with Gateway
- Using natural Language Q&A to data – Cortana

Insights And Subscriptions

- Data Navigation Paths and Data Splits
- Getting data from existing systems
- Data Refresh and LIVE Connections
- pbit and pbix: differences. Usage Options
- Quick Insights for Power BI Reports
- Quick Insights for Power BI Dashboards
- Generating Insights with Cloud Datasets
- Generating Reports with Cloud Datasets
- Using relational databases on-premises
- Using relational databases in the cloud
- Consuming a service content pack
- Creating a custom data set from a service
- Creating a content pack for your organization
- Consuming an organizational content pack
- Updating an organizational content pack
- Adding Tiles: Images, Videos, DataStream's
- Creating New Reports from Cortana, Advantages

Power BI Integration Elements

- SSRS Integration with Power BI
- SSRS Report Portal URL to Power BI Cloud
- Power BI KPI Reports Vs SSRS KPI Reports
- Conversing and Working with Mobile Reports
- Report Builder Reports to Power BI
- Generating QR Codes and Report Security
- Reporting JSON Files, Bulk Data Loads
- Creating high-density Reports in Power BI
- OLAP Data Sources in Power BI
- Using MDX Queries with Power BI Queries
- MDX SELECT and Perspective Access
- KPIs and MDX Expressions with Power BI
- MDX Queries and Filters with Power BI
- Linked Servers and T-SQL SPROCs with MDX
- YTD, PARALLELPERIOD,SCOPE, ALLMEMBERS
- WHERE, EXCEPT, RANGE, NONEMPTY
- CURRENT & EMPTY, AND / OR, LEFT / RIGHT
- Implementing Row Level Security (RLS)
- Security Roles and Role Members. Tests
- Using R for Power BI, Streaming Datasets
- Azure Connections with Power BI Desktop
- Power BI Reports using SQL Azure DBs